

5 Joint Probability Distributions

5.1 Joint Probability Distributions for Two Random Variables

Definition. Let X, Y be two random variables. With a ordered pair (X, Y) , we define a **joint probability distribution function** and denote

$$f_{XY}(x, y).$$

In discrete case, we called **joint probability mass function**; In continuous case, we called **joint probability density function**.

Property. Let X, Y be two random variables.

1. If $f_{XY}(x, y)$ is a joint probability mass function, then it satisfies the following properties:

$$(1) f_{XY}(x, y) \geq 0$$

$$(2) \sum_{x \in X} \sum_{y \in Y} f_{XY}(x, y) = 1$$

$$(3) f_{XY}(x, y) = P(X = x, Y = y) = P(X = x \text{ \& } Y = y)$$

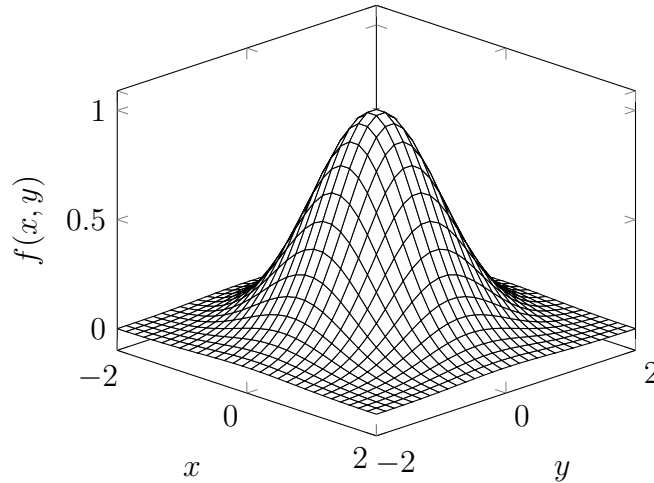
2. If $f_{XY}(x, y)$ is a joint probability density function, then it satisfies the following properties:

$$(1) f_{XY}(x, y) \geq 0 \text{ for all } x, y$$

$$(2) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = 1$$

$$(3) \text{ For any region } R \text{ of two-dimensional space,}$$

$$f_{XY}(x, y) = P((X, Y) \in R) = \iint_R f_{XY}(x, y) dx dy.$$



Definition. (Marginal Density) Let X, Y be two random variables.

1. For discrete case, the marginal(邊際) density is defined as

$$(1) f_X(x) = \sum_{y \in Y} f_{XY}(x, y)$$

$$(2) f_Y(y) = \sum_{x \in X} f_{XY}(x, y)$$

2. For continuous case, the marginal density is defined as

$$(1) f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$

$$(2) f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$

Example. Suppose $f_{XY}(x, y)$ is a joint probability mass function with two random variables $X = \{0, 1, 2\}$ and $Y = \{0, 1, 2, 3\}$. It is as shown in the table below

$Y = y$	$X = x$		
	0	1	2
0	0.15	0.1	0.05
1	0.02	0.1	0.05
2	0.02	0.03	0.2
3	0.01	0.02	0.25

Then the marginal probability distributions of X and Y are

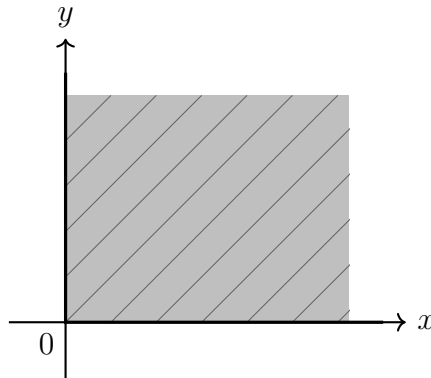
$Y = y$	$X = x$			
	0	1	2	$f_Y(y)$
0	0.15	0.1	0.05	0.3
1	0.02	0.1	0.05	0.17
2	0.02	0.03	0.2	0.25
3	0.01	0.02	0.25	0.28
$f_X(x)$	0.2	0.25	0.55	1

Example. Let $f_{XY}(x, y)$ be a joint probability density function and be defined as

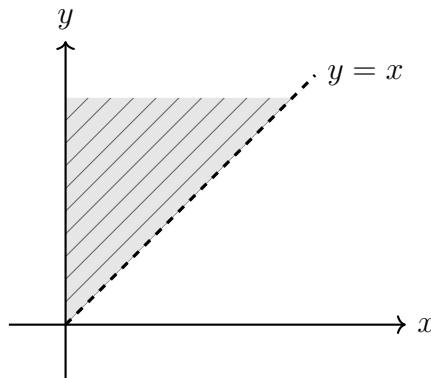
$$f_{XY}(x, y) = ke^{-0.01x-0.02y} \text{ for } 0 < x < y < \infty,$$

where $k = 6 \times 10^{-6}$. The region with nonzero probability is shaded in below. Find the $P(Y > 2000)$.

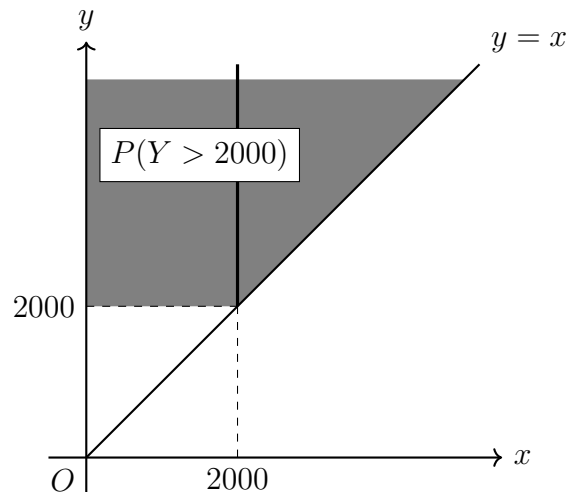
(i) **Region :** $0 < x < \infty, 0 < y < \infty$



(ii) **Region :** $0 < x < y < \infty$



Sol. The region is shaded in below.



So

$$\begin{aligned}
P(Y > 2000) &= \int_{2000}^{\infty} \int_0^y f_{XY}(x, y) dx dy \\
&= \int_{2000}^{\infty} \int_0^y k e^{-0.01x-0.02y} dx dy \\
&= k \int_{2000}^{\infty} e^{-0.02y} \left[\int_0^y e^{-0.01x} dx \right] dy \\
&= k \int_{2000}^{\infty} e^{-0.02y} \left[\frac{e^{-0.01x}}{-0.01} \right]_0^y dy \\
&= k \int_{2000}^{\infty} e^{-0.02y} \cdot 100(1 - e^{-0.01y}) dy \\
&= 100k \int_{2000}^{\infty} (e^{-0.02y} - e^{-0.03y}) dy \\
&= 100k \lim_{b \rightarrow \infty} \left[\frac{e^{-0.02y}}{-0.02} - \frac{e^{-0.03y}}{-0.03} \right]_{2000}^b \\
&= 100k \left[(0 - 0) - \left(\frac{e^{-0.02(2000)}}{-0.02} - \frac{e^{-0.03(2000)}}{-0.03} \right) \right] \\
&= 100k \left(50e^{-40} - \frac{100}{3}e^{-60} \right)
\end{aligned}$$

Substitute $k = 6 \times 10^{-6}$:

$$\begin{aligned}
P(Y > 2000) &= 6 \times 10^{-4} \left(50e^{-40} - \frac{100}{3}e^{-60} \right) \\
&= 0.03e^{-40} - 0.02e^{-60}.
\end{aligned}$$